Preliminary Multimodal Deep Learning Investigation of Tumor Immune Microenvironment Cell-Type Deconvolution for Colorectal Cancer Prognostication

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ABSTRACT

- **Deep learning** models incorporate multiple **heterogenous** modalities for high quality cancer prognostication
- **Tumor microenvironment** (TME) can be characterized by deconvolved **cell-type** proportions, which are currently untapped in **multimodal** prognostication
- **Deconvolved** cell-type proportion from **DNA Methylation**, and incorporate this information into a multimodal deep learning **prognostication** model

INTRODUCTION

- **Multimodal Machine Learning for Colorectal Cancer Analysis**
  - Predicting relevant targets (ex; diagnosis, prognosis/instantaneous risk of death, response to therapy) using input data sources collected from patients
  - Neural networks use matrices and transformations to condense numerical inputs to outputs, optimized by objective functions
  - Multimodal models — considering multiple input sources at once — shown to outperform unimodal models
  - Current methods use inputs of: histopathology, -omics (ex; DNAm, mRNA), clinical information (ex; age, gender, race)
  - Colorectal cancer is 2nd leading cause of worldwide cancer deaths; automated decision aids may improve screening/treatment
- **The Tumor Microenvironment (TME)**
  - Complex mixture of diverse cell types surround and interact with tumor tissue, forming unique environment stratified by patient characteristics; cell types often confound solid tissue analysis
  - Immune cell type proportion deconvolution algorithms operate on omics data; DNAm-based methods most accurate
  - *HiTIMED* enables highly granular TME cell type deconvolution (up to 17 cellular lineages reflecting tumor, angiogenic and immune lineages) from DNAm using hierarchical method
- **Prognostic Value of the TME in Colorectal Cancer**
  - Existing studies demonstrate connection between TME characteristics (ex; cytotoxicity) and cancer prognosis
  - Colorectal cancer treatment effects can be dependent on TME composition
  - Immune cell proportions shown to add prognostic value to omics data in integrate analyses
  - TME information may complement other modalities like histopathology imaging
  - Current lack of incorporating immune cell proportions into multimodal cancer prognostication models

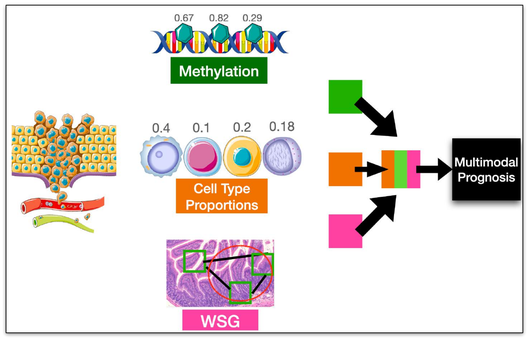

Figure 1: Workflow to include cell type proportions in multimodal prognostication

METHODS

Goal: Develop multimodal prognostication model which incorporates novel combination of histopathology, DNAm, and immune cell proportions, and assess capability relative to existing methods

- **Data Acquisition and Preprocessing**
  - Matched histopathological whole slide imaging (WSI) and bulk DNAm profiling collected from 414 TCGA colorectal cancer patients
  - Data split into train, test, validation sets (80/10/10) at the patient-level
  - DNAm profiles preprocessed to remove SNP-linked and sex-linked CpGs using *sesame* R package
  - *HiTIMED* applied to deconvolve proportions for 17 immune cell types from each bulk DNAm profile
  - WSI preprocessed by breaking into patches connected as graph, where nodes are patch image representations from *ResNet* CNN model pretrained on *ImageNet* dataset (Figure 2)
  - Each patient paired with single WSI, DNAm profile, and cell proportions profile
- **Model Development and Validation**
  - Unimodal encoders (graph convolutional network aka GCN for WSI, multilayer perception aka MLP for cell proportions, variational autoencoder aka VAE + MLP for DNAm) pretrained using survival objective
  - Multimodal model (*CRCMulti*) (Figure 3) designed where embeddings extracted from unimodal arms, fused via *Pathomic Fusion* method, subsequent MLP applied for final risk output
  - Pretrained unimodal encoder weights transferred to multimodal model at initialization stage
  - Training procedure was supervised using Cox Loss, model checkpoint selected based on validation-set loss
  - Final performances were assessed using concordance index with (c-index) 1000 sample nonparametric bootstrapping on heldout testing dataset split
  - *CRCMulti* compared versus unimodal-only models, and other reported TCGA multimodal colorectal prognostication methods

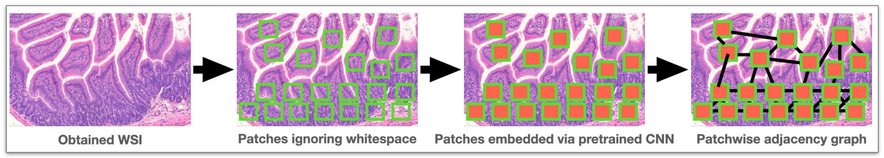

Figure 2: Preprocessing workflow for histopathology imaging data from obtaining WSI to creating representative graph

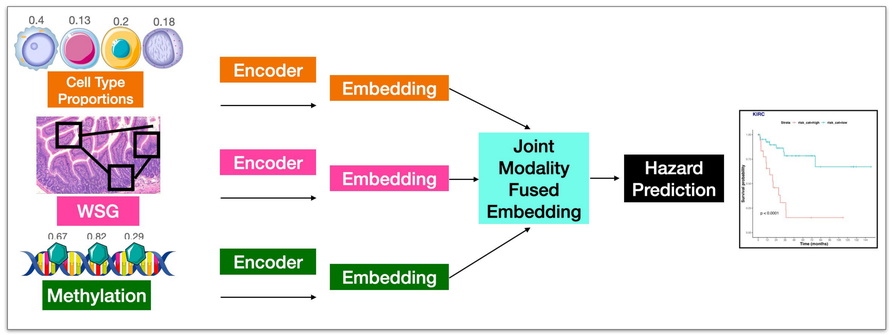

Figure 3: Overview of multimodal colorectal cancer prognostication model (*CRCMulti*), including unimodal embedding, fused embedding creation, and final output hazard/prognosis prediction

RESULTS

Model	Test C-Index
SoloHiTIMED	0.623 ± 0.003
SoloDNAm	0.615 ± 0.001
SoloWSI	0.593 ± 0.003
CRCMulti	**0.672 ± 0.002**

Table 1: Prognostic power of unimodal models (*Solo-*) and multimodal model (*CRCMulti*)

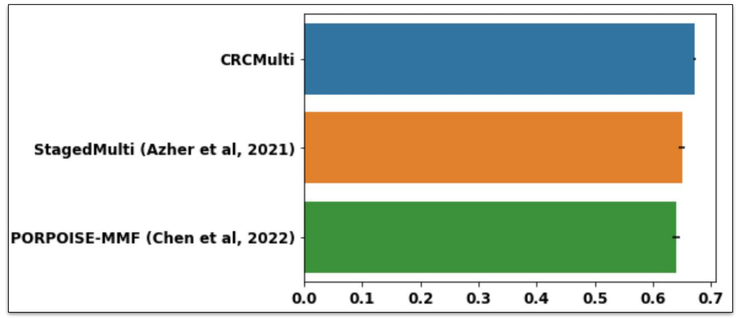

Figure 4: Comparison of *CRCMulti* with existing reported multimodal model performance metrics

CONCLUSION

- **Potential Clinical Value of TME**
  - TME information is an independent biomarker of colorectal cancer prognosis
  - Immune cell proportions can be leveraged to develop capable multimodal prognostication methods
  - Integration of TME with existing data sources can improve predictive power
- **Limitations**:
  - Lack of using additional external validation cohorts
  - Model training optimization via coarse hyperparameter searches
  - Interpretability of features extracted from TME data is unclear
- **Future Directions:**
  - Curate larger cohort and train/validate model with more datasets and cancer types
  - Include additional data modalities (ex; clinical, mRNA) and investigate combinations where TME may/may not enhance performance
  - Systematic ablation studies to determine when specifically TME brings additional prognostic power
  - Apply interpretation techniques to elucidate model reasoning using TME data
- **Data and Code Availability:**
  - Data available on reasonable request, privacy/ethical restrictions.
- **Funding:**
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References: Available at tinyurl.com/tme-multimodal-pbs-poster-refs